

Is GDP per Capita a Good Predictor of National Well-Being? Income versus World Regions*

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Abstract

Across countries, national well-being is commonly described as an increasing function of log GDP per capita. I assess how much of national well-being GDP per capita actually predicts, and benchmark it against a simple alternative: the country's world region. Using all waves of the World Values Survey and the European Values Study (1981–2023, 112 countries), eight well-being indicators and seven income measures, I find that log GDP per capita explains on average 14% of the cross-country variance of well-being, and nothing of the share of very happy people. Region is a better predictor in 85% of 1,064 specifications and, out of sample, for every indicator. This advantage is driven by Latin America (happier than its income predicts) and the former Eastern Bloc (unhappier), and it does not hold in the Gallup World

*Data and code: https://github.com/bixiou/wellbeing_gdp_region [TODO: public repository (e.g. OSF); Gallup data cannot be redistributed]. I thank Claudia Senik and participants in the 2024 STATEC Well-being Conference for helpful comments, Armon Rezai and the Vienna University of Economics and Business (WU Wien) for access to the Gallup World Poll data, and Julieta Toffoli for research assistance. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Declaration of interest: none.

Poll, where income predicts life evaluations as well as region. Within countries, however, well-being rises with GDP over time. Whether GDP is a poor predictor of national well-being thus depends on the data and on a few large regional deviations.

Keywords: subjective well-being; life satisfaction; happiness; GDP; world regions; Easterlin paradox.

JEL codes: I31; O15; O57.

1 Introduction

Which country is the happiest? The answer most often heard, based on the World Happiness Report, is a Scandinavian, high-income country. Behind this answer lies one of the most cited stylized facts of well-being research: across countries, average life evaluations increase with the logarithm of GDP per capita (Deaton, 2008; Sacks et al., 2012; Stevenson & Wolfers, 2008). Using the Gallup World Poll, Deaton (2008) finds a correlation above 0.8 between average life evaluation and log GDP per capita; using the World Values Survey (WVS), Inglehart & Klingemann (2000) reported an R^2 of about one half for an index combining happiness and life satisfaction. These correlations are frequently invoked to support the view that economic growth is a reliable route to well-being.

This paper asks how much of national well-being GDP per capita actually predicts, and whether it predicts it better than a very simple alternative, the country's world region. Region is a natural benchmark: previous work has documented that Latin American countries report higher well-being than their income would predict (Graham & Lora, 2009; Inglehart et al., 2008; Rojas, 2016) and former communist countries lower well-being (Guriev & Zhuravskaya, 2009), and that cultural factors shape national well-being (Diener & Suh, 2000; Diener et al., 2003; Exton et al., 2015; Senik, 2014). But the relative predictive power of income and region has not, to my knowledge, been systematically compared.

I use all seven waves of the WVS, complemented with the 2017 wave of the European Values Study (342 country-years), eight indicators of national well-being based on the happiness and life-satisfaction questions, and seven income measures (log GDP per capita at PPP or market exchange rates, income sextiles,

and income clusters). Since the ranking of groups can depend on the choice of indicator (Bond & Lang, 2019), I consider all of them. I compare the variance explained by income and by region using the Shapley (LMG) decomposition of R^2 (Grömping, 2007; Lindeman et al., 1980), adjusted R^2 , and leave-one-country-out cross-validation, which directly measures predictive capacity. I then examine the robustness of the results across waves, samples, classifications of regions, and datasets, using the Gallup World Poll and the World Happiness Report.

Four findings emerge. First, in the WVS, GDP per capita is a weak predictor of national well-being: log GDP per capita explains on average 14% of the variance of the eight indicators, and nothing of the share of very happy people. Second, region is a better predictor than income in 85% of specifications and, out of sample, for every indicator. Third, this conclusion must be qualified: the advantage of region is driven by Latin America and the former Eastern Bloc, it weakens with coarser classifications, and it does not hold in Gallup, where log GDP explains 61% of the variance of the mean ladder against 48% for region. Fourth, within countries, well-being increases with GDP over time.

The paper contributes to the literature on income and well-being across countries (Deaton, 2008; Easterlin, 1974; Easterlin et al., 2010; Jebb et al., 2018; Stevenson & Wolfers, 2008) by quantifying the predictive capacity of income out of sample, benchmarking it against regions, and showing that the strength of the cross-country relationship depends on the dataset. A companion paper [TODO: cite companion paper] uses a survey experiment to show that the discrepancy between Gallup and the WVS is not due to question wording.

Section 2 presents the data, Section 3 the results, Section 4 discusses them, and Section 5 concludes.

2 Data

2.1 World Values Survey

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (Haerpfer et al., 2022), which pools the seven waves conducted between 1981 and 2022, and complete it with the Joint EVS/WVS 2017–2022 dataset (EVS/WVS, 2024), which adds

the 36 national surveys of the 2017 wave of the European Values Study (EVS, conducted in 2017–2021) and 2 recent WVS surveys (India 2023, Uzbekistan 2022). The EVS asks the same well-being questions, and the combination of the two surveys is standard practice (the so-called Integrated Values Surveys). The EVS and the WVS are distinct surveys: when both surveyed the same country the same year (3 cases), I keep both surveys as separate observations. For brevity, I refer to the combined data as the WVS: 501,911 respondents in 342 country-years and 112 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the WVS in the United States in 2017 and in Japan in 2019, or the EVS in Switzerland in 2017). All statistics use the survey weights provided by the WVS and the EVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: “Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*.”
- *Life satisfaction*: “All things considered, how satisfied are you with your life as a whole these days?”, answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.4% for satisfaction and 1.7% for happiness on average) and uncorrelated with GDP per capita (correlation: -0.09); they are excluded.

Indicators of national well-being. Because there is no consensus on how to aggregate ordinal answers into a national indicator, and because the ranking of countries can depend on this choice (Bond & Lang, 2019), I compute eight indicators for each country-year:

- *Happy*: share answering *Quite* or *Very happy*;
- *Very Happy*: share answering *Very happy*;
- *Very Unhappy*: share answering *Not at all happy*;
- *V. Happy – V. Unhappy*: difference between the two previous shares;

- *Happiness (mean)*: mean happiness, with answers coded +3, +1, -1, -3;
- *Satisfaction (mean)*: mean life satisfaction;
- *Satisfied*: share answering 6 to 10 to the life-satisfaction question;
- *Happy + Satisfied*: average of *Happy* and *Satisfied*, the index used by Inglehart & Klingemann (2000).

For the life-satisfaction question, I also compute the shares answering 8–10, 9–10, 10, 1–4 and 1–2, which describe the tails of the distribution and are used for comparisons with Gallup. Finally, I compute the share of people whose satisfaction is below 60% of the national mean, by analogy with relative poverty lines. The indicators are positively but imperfectly correlated: across country-years, the correlation between *Happy* and *Satisfied* is 0.72, and between *Very Happy* and *Satisfaction (mean)* it is only 0.60.

2.2 Gallup World Poll and World Happiness Report

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (Gallup, 2022). Its life-evaluation question is the Cantril ladder (Cantril, 1965): “Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?”

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2,351 country-years, 168 countries, 2006–2023), from which I compute the same satisfaction indicators as for the WVS: the mean and the shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2. These distributions are unweighted. To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (World Happiness Report, 2026), which average Gallup data over three years up to 2023–2025.

2.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators (World Bank, 2026). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are filled in automatically, in three steps: (i) PPP series, which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita; (ii) for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook (International Monetary Fund, 2026), whose values are in current dollars; I convert them into the constant dollars of the World Bank series by multiplying them by the ratio, in the same year, between the World Bank and the IMF values for the United States (a U.S. price deflator, since PPP and market-exchange-rate values coincide for the United States); (iii) remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for Northern Ireland). A robustness check excludes all imputed values.

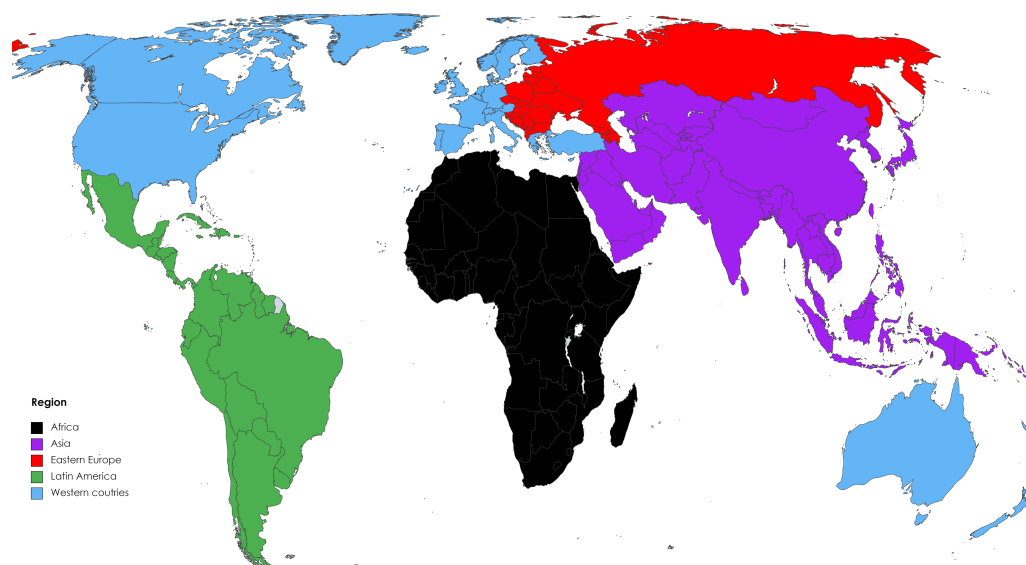
Region is a categorical variable, which can capture non-linear patterns that a single continuous variable cannot. To compare income and region on an equal footing, I also use categorical income indicators: income sextiles and income clusters obtained by k -means clustering of log GDP per capita, with $k = 5, 6$ or 7 . The clusters let the data choose the thresholds between income groups and allow for non-linear relationships, at the cost of more parameters than the log-linear specification. Sextiles and clusters are recomputed in each estimation sample: for instance, when the analysis is restricted to wave 3, the income groups are defined from the country-years surveyed in wave 3 only.

2.4 Regions

My main classification groups countries into the five UN regional groups (Figure 1): *Africa*, *Asia* (the Asia-Pacific group, which includes the Middle East and Central Asia), *Eastern Europe* (the former Eastern Bloc, excluding Central Asia), *Latin America* (including the Caribbean), and *Western* countries (the “Western European and Others” group: Western Europe, North America, Australia, New

Zealand, and Turkey). It departs from the UN groups for two countries only: Israel, a member of the Western group since 2000 because it was denied access to the Asia-Pacific group, is classified in Asia, and Cyprus, a member of the Asia-Pacific group, is classified with Western countries. Territories that are not UN members (e.g. Taiwan, Hong Kong, Puerto Rico, Kosovo) are assigned to the group of their neighbors. For robustness, I use five alternative classifications: the exact UN regional groups (with Israel in the Western group and Cyprus in Asia); a six-region variant separating the Middle East and grouping Central Asia with the former Eastern Bloc; the seven World Bank regions; the five continents; and the UN geoscheme sub-regions (about twenty groups).

Figure 1: Main classification of countries into five world regions



3 Results

3.1 Empirical strategy

For each well-being indicator and each income indicator, I estimate three linear regressions across country-years i :

$$well-being_i = \alpha_1 + \beta_1 income_i + u_i \quad (1)$$

$$well-being_i = \alpha_2 + \gamma_2 region_i + e_i \quad (2)$$

$$well-being_i = \alpha_3 + \beta_3 income_i + \gamma_3 region_i + \varepsilon_i \quad (3)$$

where $region_i$ denotes the vector of region dummies (so that γ_2 and γ_3 are vectors of coefficients). Denoting R_1^2 , R_2^2 and R_3^2 their coefficients of determination, the share of the variance explained by income alone is R_1^2 . To compare income and region, I decompose R_3^2 between the two regressors following Lindeman et al. (1980) (LMG), i.e. using the Shapley value (Grömping, 2007). With two (groups of) regressors, the LMG contribution of income is the average of its contribution when entered first (R_1^2) and when entered second ($R_3^2 - R_2^2$). The *share of explained variance due to income* is therefore

$$s = \frac{R_1^2 + (R_3^2 - R_2^2)}{2 R_3^2}, \quad (4)$$

and region is the better predictor whenever $s < 50\%$.

One feature of this comparison could favor region: region has four degrees of freedom while log GDP has one. I address this concern in three ways. First, I use categorical income variables (sextiles and clusters) with as many or more categories as regions. Second, I report the share of explained variance due to income computed with adjusted R^2 , which penalizes additional regressors. I keep the unadjusted R^2 as the benchmark because it is the statistic used in the literature, and because the adjustment only reinforces the conclusions: with more than 300 observations, the adjustment is small, and with adjusted R^2 region is the better predictor for every indicator and income measure of Table 2. Third, and more directly relevant to the question of *predictive* capacity, I compute leave-one-country-out cross-validated R^2 : each country's observations are predicted by a

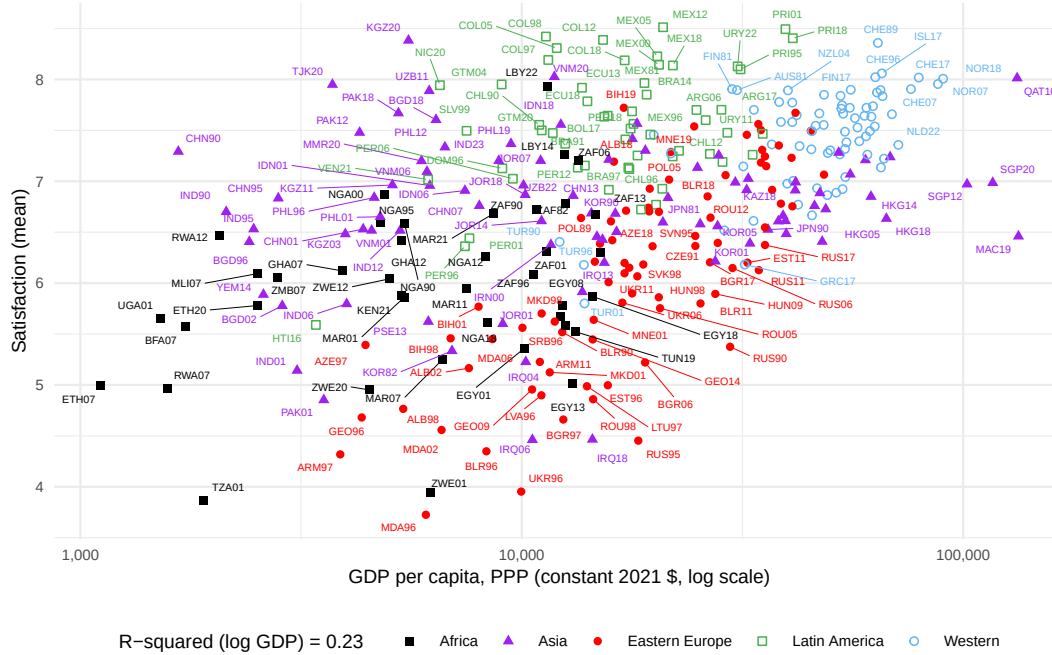
model estimated on all other countries, and the cross-validated R^2 measures the share of the variance of well-being that the model correctly predicts for countries it was not estimated on. Unlike the in-sample R^2 , it decreases when a model has superfluous parameters, and it is negative when a model predicts worse than the sample mean. Besides, as results could depend on how countries are grouped, I report results for five alternative classifications. Other robustness checks include each WVS wave separately, excluding the EVS surveys, weighting countries by their population, keeping only the last observation of each country, excluding the pandemic years 2020–2021, excluding imputed GDP values, using the previous vintage of GDP data (PPP in constant 2017 dollars), and excluding Latin America and Eastern Europe.

3.2 Income explains a small share of national well-being in the WVS

Figure 2 plots mean life satisfaction against GDP per capita for all WVS country-years. The relationship is positive, but the dispersion at any given income level is large: countries with a GDP per capita around 10,000 dollars span most of the range of observed life satisfaction, from the former Soviet republics at the bottom to Latin American countries at the top. Log GDP per capita explains 23% of the variance of mean satisfaction.

Table 1 generalizes this result to all indicators and income measures. On average over the eight indicators, log GDP per capita at PPP explains 14% of the variance, and the best-performing income measure (Cluster k=7 nominal) explains 20%. The indicators are positively but imperfectly correlated (Appendix Table A1), and satisfaction-based indicators are better explained than happiness-based ones: log GDP explains 4% of the variance of mean happiness and essentially none of the share of *Very happy* people, whose slope with respect to log GDP is statistically indistinguishable from zero (slope 0.002, $p = .948$; Appendix Table A2). This confirms the observation of Deaton (2008) and Inglehart et al. (2008) that income is more closely related to life evaluation than to happiness. By contrast, region alone explains 33% of the variance on average (last column).

Figure 2: Mean life satisfaction vs. GDP per capita (PPP), WVS/EVS 1981–2023



Note: Each point is a country-year (label: ISO code and last two digits of the year). The legend gives the R^2 of the regression on log GDP per capita.

Table 1: Share of the variance of national well-being explained by income alone (R^2), WVS/EVS 1981–2023

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster				Region
	PPP	nominal		k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal	
Happy	0.14	0.20	0.20	0.19	0.20	0.20	0.23	0.30
Very Happy	0.00	0.00	0.03	0.03	0.04	0.05	0.04	0.33
Very Unhappy	0.07	0.10	0.12	0.10	0.13	0.13	0.14	0.14
V. Happy – V. Unhappy	0.00	0.01	0.04	0.04	0.04	0.05	0.05	0.33
Happiness (mean)	0.04	0.08	0.10	0.08	0.09	0.10	0.11	0.33
Satisfaction (mean)	0.23	0.29	0.23	0.21	0.22	0.23	0.29	0.41
Satisfied	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.39
Happy + Satisfied	0.27	0.35	0.30	0.28	0.30	0.30	0.35	0.38
Mean	0.14	0.18	0.17	0.15	0.17	0.17	0.20	0.33
Max	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.41

Note: R^2 of the regression of each well-being indicator (rows) on each income indicator (columns), over 342 WVS/EVS country-years (a few with missing GDP are excluded). The last column gives the R^2 of the regression on region dummies. The replication files include an Excel version of this table with the R^2 of all region classifications (tables/main/r2_income_all_regions.xlsx).

3.3 Region predicts national well-being better than income in the WVS

Table 2 reports the share of explained variance due to income, s , for each indicator and income measure. It is below 50% in all cases: on average, the best income measure accounts for 33% of the variance explained jointly by income and region. The first split of a regression tree using log GDP and region as candidate variables is on region for 8 out of 8 indicators.

Table 2: Share of explained variance due to income rather than region (LMG), WVS/EVS 1981–2023

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		$k=5$ PPP	$k=6$ PPP	$k=7$ PPP	$k=7$ nominal
Happy	0.29	0.36	0.38	0.35	0.37	0.38	0.41
Very Happy	0.00	0.01	0.07	0.06	0.08	0.10	0.10
Very Unhappy	0.28	0.38	0.43	0.40	0.47	0.47	0.50
V. Happy – V. Unhappy	0.01	0.02	0.09	0.07	0.09	0.11	0.11
Happiness (mean)	0.09	0.13	0.18	0.14	0.17	0.19	0.19
Satisfaction (mean)	0.32	0.38	0.32	0.29	0.31	0.32	0.38
Satisfied	0.43	0.49	0.43	0.41	0.43	0.44	0.49
Happy + Satisfied	0.39	0.46	0.42	0.39	0.41	0.42	0.47
Mean	0.23	0.28	0.29	0.26	0.29	0.30	0.33
Max	0.43	0.49	0.43	0.41	0.47	0.47	0.50

Note: Share of the R^2 of Equation (3) attributed to income by the LMG (Shapley) decomposition, Equation (4). Values below 0.5 indicate that region is the better predictor.

The result is not an artifact of the larger number of parameters of the region model. Out of sample, region predicts better than income for every indicator (Table 3): for mean satisfaction, the leave-one-country-out R^2 is 0.37 with region against 0.21 with log GDP. For several happiness indicators, income has no out-of-sample predictive power at all (negative cross-validated R^2).

Table 4 summarizes 1,064 estimates (indicator $\times$ income measure $\times$ specification) using the WVS. Region is the better predictor in 85% of them (76% when restricting to the best income measure). The result holds in each wave except wave 4 (1999–2004, 40 countries), where income and region perform similarly; when countries are weighted by population; when keeping one observation per country; without the pandemic years; without imputed GDP values; with the previous vintage of GDP data; without the EVS surveys; and with the exact UN groups, the six-region or the UN sub-region classifications.

Table 3: Out-of-sample predictive capacity: leave-one-country-out cross-validated R^2

Data	Indicator	Obs.	log GDP PPP	Best income variable	Region	log GDP PPP + Region
WVS, all waves	Happy	340	0.12	0.18	0.26	0.32
WVS, all waves	Very Happy	340	−0.03	−0.04	0.27	0.27
WVS, all waves	Very Unhappy	340	0.06	0.10	0.10	0.11
WVS, all waves	V. Happy – V. Unhappy	340	−0.02	−0.02	0.27	0.26
WVS, all waves	Happiness (mean)	340	0.02	0.05	0.29	0.30
WVS, all waves	Satisfaction (mean)	340	0.21	0.23	0.37	0.46
WVS, all waves	Satisfied	340	0.30	0.34	0.36	0.50
WVS, all waves	Happy + Satisfied	338	0.26	0.31	0.35	0.47
WVS, last obs. per country	Happy	115	0.05	0.03	0.11	0.12
WVS, last obs. per country	Very Happy	115	0.01	−0.05	0.17	0.16
WVS, last obs. per country	Very Unhappy	115	0.04	0.06	0.01	0.02
WVS, last obs. per country	V. Happy – V. Unhappy	115	−0.01	−0.08	0.16	0.15
WVS, last obs. per country	Happiness (mean)	115	−0.04	−0.09	0.12	0.10
WVS, last obs. per country	Satisfaction (mean)	115	0.18	0.19	0.30	0.33
WVS, last obs. per country	Satisfied	115	0.29	0.31	0.31	0.38
WVS, last obs. per country	Happy + Satisfied	115	0.24	0.25	0.28	0.34
Gallup, all years (2006–23)	Satisfaction (mean)	2344	0.61	0.66	0.49	0.68
Gallup, all years (2006–23)	Satisfied	2344	0.65	0.72	0.52	0.72
Gallup, all years (2006–23)	Unsatisfied (0/1–4)	2344	0.63	0.65	0.48	0.68
Gallup, last obs. per country	Satisfaction (mean)	167	0.60	0.58	0.45	0.64
Gallup, last obs. per country	Satisfied	167	0.65	0.68	0.52	0.70
Gallup, last obs. per country	Unsatisfied (0/1–4)	167	0.65	0.60	0.48	0.67
WHR, 2023–25 average	Satisfaction (mean)	147	0.62	0.60	0.52	0.69

Note: Each country’s observations are predicted with a model estimated on all other countries. Cross-validated $R^2 = 1 - \sum(y - \hat{y}_{\text{out-of-sample}})^2 / \sum(y - \bar{y})^2$; it can be negative when a model predicts worse than the overall mean.

Table 4: Robustness: income vs. region across specifications and datasets

Specification	Obs.	Countries	R^2 income log PPP	R^2 best	R^2 region	Share of explained variance due to income log PPP	best	adj. R^2	Region better (share of cases)
WVS, all waves	340	112	0.14	0.20	0.33	0.23	0.33	0.27	0.98
WVS, waves 1–2 (1981–91)	26	21	0.07	0.43	0.58	0.09	0.40	0.10	1.00
WVS, wave 3 (1995–99)	55	55	0.16	0.33	0.65	0.14	0.28	0.18	1.00
WVS, wave 4 (1999–2004)	40	40	0.19	0.31	0.33	0.27	0.47	0.47	0.43
WVS, wave 5 (2004–09)	58	58	0.21	0.35	0.49	0.22	0.38	0.23	0.98
WVS, wave 6 (2010–16)	60	60	0.08	0.22	0.27	0.14	0.41	0.19	0.96
WVS, wave 7 (2017–23)	102	92	0.09	0.17	0.28	0.21	0.35	0.27	1.00
WVS, wave 7 w/o 2020–21	80	72	0.08	0.19	0.29	0.19	0.38	0.27	0.95
WVS, population-weighted	340	112	0.10	0.19	0.21	0.25	0.44	0.32	0.77
WVS, last obs. per country	115	112	0.13	0.19	0.25	0.26	0.39	0.30	0.88
WVS, no GDP imputation	321	108	0.13	0.20	0.32	0.22	0.33	0.27	0.98
WVS, GDP PPP 2017 \$ (old)	340	112	0.14	0.20	0.33	0.24	0.33	0.28	0.98
WVS, without EVS surveys	304	108	0.12	0.19	0.37	0.18	0.28	0.22	1.00
WVS, w/o Latin Am. & E. Europe	201	72	0.18	0.28	0.20	0.35	0.68	0.54	0.27
WVS, 6 regions	340	112	0.14	0.20	0.32	0.22	0.33	0.27	1.00
WVS, World Bank regions (7)	340	112	0.14	0.20	0.19	0.37	0.50	0.44	0.55
WVS, continents (5)	340	112	0.14	0.20	0.17	0.40	0.53	0.47	0.48
WVS, exact UN regional groups	340	112	0.14	0.20	0.33	0.23	0.33	0.27	1.00
WVS, UN sub-regions	340	112	0.14	0.20	0.37	0.24	0.32	0.29	0.96
Gallup, all years (2006–23)	2344	167	0.42	0.48	0.42	0.46	0.51	0.49	0.31
Gallup, last obs. per country	167	167	0.42	0.44	0.41	0.44	0.48	0.45	0.37
Gallup, last obs., WVS countries	109	109	0.44	0.47	0.43	0.46	0.49	0.47	0.35
Gallup, last obs., 6 regions	167	167	0.42	0.44	0.42	0.43	0.46	0.44	0.41
Gallup, last obs., WB regions	167	167	0.42	0.44	0.38	0.47	0.50	0.48	0.29
WHR, 2023–25 average	147	147	0.63	0.63	0.55	0.56	0.56	0.56	0.00

Note: Averages over well-being indicators (8 for the WVS; 7 satisfaction indicators for Gallup; mean ladder for the WHR). “Best” refers to the income measure with the highest average R^2 in the main WVS specification (Cluster k=7 nominal). “Region better” is the share of indicator $\times$ income-measure combinations for which $s < 0.5$.

3.4 Where the region advantage comes from

Three results qualify this conclusion. First, the advantage of region is driven by two well-known “anomalies”: Latin America, happier than its income would predict (Graham & Lora, 2009; Rojas, 2016), and the former Eastern Bloc, unhappier (Guriev & Zhuravskaya, 2009). Excluding these two regions, region is the better predictor in only 27% of cases. Second, the result depends on the granularity and design of the classification: with continents or World Bank regions, which do not isolate Latin America or the former Eastern Bloc, region is better in 48% and 55% of cases, respectively. Third, and most importantly, the conclusion does not carry over to Gallup data. In Gallup’s latest observation for each of 168 countries, log GDP explains 61% of the variance of the mean ladder, against 48% for region, and income accounts for 60% of the explained variance (Appendix Table A3). Over all Gallup specifications, region is the better predictor in only 34% of cases, mostly for indicators of the upper tail of the distribution (shares answering 9–10 or 10), which are almost unrelated to income. With the latest WHR data (2023–2025), log GDP explains 63% of the variance of the ladder against 55% for region.

The relation between GDP and well-being is thus much weaker in the WVS than in Gallup. The same holds when restricting both datasets to the 187 country-years observed by both surveys: the R^2 of mean life evaluation on log GDP is 0.53 in Gallup and 0.16 in the WVS. Whether “GDP is a poor predictor of national well-being” therefore depends on which of the two datasets one trusts.

3.5 Other correlates and within-country evidence

What do regions capture? Appendix Table A4 decomposes the variance of mean satisfaction among income, region and one additional country-level variable from the WVS. The average perceived freedom of choice and control over one’s life (“how much freedom of choice and control you feel you have over the way your life turns out”) alone explains 63% of the variance of mean satisfaction; in a Shapley decomposition with income and region, it accounts for 42 percentage points, against 21 for region and 11 for income. This variable is itself a subjective evaluation, so this correlation should not be read causally, but it suggests that non-material dimensions of life are central to regional differences (Inglehart et al.,

2008). Tolerance of homosexuality (24%), the importance of God (1%) and GDP growth over the previous five years (2%) explain much less.

Finally, the cross-sectional relation should not be confused with the relation over time. Within countries (country fixed effects, 312 observations from countries surveyed at least twice), mean satisfaction increases with log GDP per capita, with a slope of 2.31 per unit of $\log_{10}$ GDP (i.e. per tenfold increase), larger than the cross-sectional slope (1.14), and even the share of *Very happy* people increases (slope 0.21, $p < .001$). Income growth explains 0.34 of the within-country variance (Appendix Table A5). These within-country associations are consistent with Stevenson & Wolfers (2008) and Sacks et al. (2012) rather than with a strict version of the Easterlin paradox (Easterlin, 1974; Easterlin et al., 2010), although they may reflect common shocks (e.g. transition recessions) rather than the effect of income per se.

3.6 What are the happiest countries?

Rankings of countries depend on the indicator. Taking, for each indicator and each wave (and all waves combined), the happiest country-year, Switzerland appears most often (10 times out of 64), followed by Mexico (9) and Kyrgyzstan (6) (Appendix Table A6). The happiest country-years are Western (25), Latin American (19), Asian (15) or African (5). The richest countries are thus not necessarily the happiest: the link between income and well-being seems to be that rich countries rarely experience low well-being, rather than that they reach the highest levels.

4 Discussion

Several caveats apply to the comparison between income and region. First, region is not a mechanism: it bundles culture, history, institutions, social ties, and possibly cross-cultural differences in how people use response scales (Chen et al., 1995; Diener et al., 2003; Exton et al., 2015). If regional differences partly reflect response styles, then region predicts *measured* well-being without necessarily predicting *experienced* well-being. Existing evidence suggests that self-reported well-being carries information about true well-being (Kaiser & Oswald, 2022; Sandvik

et al., 1993), and studies exploiting migrants suggest that national differences partly reflect a cultural component of well-being rather than mere translation artifacts (Exton et al., 2015; Senik, 2014), but the present analysis cannot separate these interpretations. Anchoring vignettes (Kapteyn et al., 2010) or measures of emotional well-being would help.

Second, the advantage of region rests largely on Latin America and the former Eastern Bloc. These are well-documented deviations from the income gradient, but they are not independent of the classification: grouping countries into continents or World Bank regions, which do not isolate these groups, weakens the result. The fairest summary is that GDP per capita leaves much of the cross-country variation unexplained, and that a few large regional deviations explain more of it than income does in the WVS.

Third, the analysis pools country-years that are not independent, and R^2 decompositions carry no information on causality. The within-country evidence shows that growth is associated with rising life satisfaction, so the weak cross-sectional relation does not imply that income is irrelevant for well-being.

Fourth, and most importantly, the WVS and Gallup disagree on the strength of the income gradient. In Gallup, income predicts life evaluations about as well as region, or better. Which dataset should be trusted? Gallup has a more standardized design, and WVS samples are less representative in poorer countries: they over-represent tertiary-educated adults by a median factor of 1.98 in countries below 10,000 dollars of GDP per capita, against 1.05 above 30,000 dollars. Yet reweighting WVS samples to the population share of tertiary-educated adults barely changes their income gradient. The companion paper [TODO: cite] investigates this discrepancy with a survey experiment in ten high-income countries. It finds that question wording (Cantril ladder vs. life satisfaction) shifts the level of answers but hardly the income gradient: asked to the same sample, both questions yield a gradient close to Gallup's, so the discrepancy reflects differences in samples, interview modes and context. In the absence of evidence on which dataset better measures national well-being in low-income countries, where the two disagree most, conclusions about the global income–well-being relationship should be reported for both.

5 Conclusion

GDP per capita is a weaker predictor of national well-being than is often assumed, but how weak depends on the data. In the WVS, it explains a small share of the cross-country variance of well-being, less than a country's world region, and essentially nothing of the share of very happy people. The richest countries are not necessarily the happiest: rich countries rarely experience low well-being, but many countries at intermediate income levels, in particular in Latin America, report well-being as high as the richest. In the Gallup World Poll, however, income predicts life evaluations as well as region, and within countries well-being rises with GDP over time.

These results suggest that absolute income is less determining for national well-being than the cross-country correlations often cited imply, and that non-material dimensions associated with regions—social ties, perceived freedom, history, culture—deserve more attention, both in research on the mechanisms behind regional differences and in policy evaluation (Frijters et al., 2020; Stiglitz et al., 2009). Future research should test whether emotional well-being is also better predicted by region than by income, and use anchoring vignettes to separate true differences in well-being from differences in response styles.

Declaration of generative AI and AI-assisted technologies in the writing process

[TODO: During the preparation of this work the author used Claude (Anthropic) to write analysis code and draft parts of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. (Adapt to actual use.)]

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A Additional results

Table A1: Correlation between national well-being indicators (WVS country-years)

	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
Happy	1.00	0.60	–0.73	0.70	0.89	0.73	0.72	0.90
Very Happy	0.60	1.00	–0.37	0.98	0.89	0.60	0.51	0.59
Very Unhappy	–0.73	–0.37	1.00	–0.54	–0.67	–0.58	–0.57	–0.69
V. Happy – V. Unhappy	0.70	0.98	–0.54	1.00	0.94	0.66	0.58	0.67
Happiness (mean)	0.89	0.89	–0.67	0.94	1.00	0.75	0.69	0.83
Satisfaction (mean)	0.73	0.60	–0.58	0.66	0.75	1.00	0.96	0.93
Satisfied	0.72	0.51	–0.57	0.58	0.69	0.96	1.00	0.95
Happy + Satisfied	0.90	0.59	–0.69	0.67	0.83	0.93	0.95	1.00

Table A2: Slope of national well-being indicators with respect to $\log_{10}$ GDP per capita (PPP)

Data	Indicator	Slope	s.e.	<i>p</i> -value	R^2	Obs.
WVS	Happy	0.109	0.019	0.000	0.144	340
WVS	Very Happy	0.002	0.025	0.948	0.000	340
WVS	Very Unhappy	–0.022	0.004	0.000	0.072	340
WVS	V. Happy – V. Unhappy	0.023	0.026	0.379	0.004	340
WVS	Happiness (mean)	0.264	0.083	0.002	0.044	340
WVS	Satisfaction (mean)	1.143	0.142	0.000	0.233	340
WVS	Satisfied	0.236	0.024	0.000	0.316	340
WVS	Happy + Satisfied	0.173	0.020	0.000	0.275	338
Gallup	Satisfied	0.372	0.017	0.000	0.655	2344
Gallup	Very satisfied (8–10)	0.218	0.018	0.000	0.442	2344
Gallup	Extremely satisfied (9–10)	0.065	0.009	0.000	0.178	2344
Gallup	Completely satisfied (10)	0.012	0.005	0.008	0.015	2344
Gallup	Unsatisfied (0/1–4)	–0.304	0.012	0.000	0.639	2344
Gallup	Dissatisfied (0/1–2)	–0.135	0.009	0.000	0.432	2344
Gallup	Satisfaction (mean)	1.779	0.083	0.000	0.611	2344

Note: OLS across country-years; standard errors clustered by country. Gallup: 2006–2023.

Table A3: Share of explained variance due to income rather than region (LMG), Gallup, last observation per country

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal
Satisfied	0.58	0.61	0.59	0.58	0.60	0.59	0.60
Very satisfied (8–10)	0.44	0.51	0.49	0.47	0.49	0.51	0.53
Extremely satisfied (9–10)	0.16	0.25	0.24	0.27	0.22	0.30	0.28
Completely satisfied (10)	0.02	0.02	0.08	0.09	0.06	0.08	0.08
Unsatisfied (0/1–4)	0.61	0.60	0.60	0.59	0.61	0.60	0.59
Dissatisfied (0/1–2)	0.66	0.63	0.67	0.66	0.67	0.67	0.65
Satisfaction (mean)	0.60	0.62	0.59	0.58	0.60	0.59	0.60
Mean	0.44	0.46	0.47	0.46	0.47	0.48	0.48
Max	0.66	0.63	0.67	0.66	0.67	0.67	0.65

Note: 168 countries, last Gallup observation available (2006–2023). Same definitions as Table 2.

Table A4: Other country-level correlates of national well-being: Shapley decomposition of R^2 (WVS)

Indicator	Other variable	Obs.	R^2 other alone	Shapley decomposition of R^2			
				Income	Region	Other	Total
Satisfaction (mean)	Freedom of choice	334	0.63	0.11	0.21	0.42	0.74
Satisfaction (mean)	Importance of God	329	0.01	0.17	0.33	0.02	0.51
Satisfaction (mean)	Tolerance (homosexuality)	326	0.24	0.12	0.30	0.10	0.52
Satisfaction (mean)	Importance of democracy	216	0.08	0.13	0.32	0.02	0.47
Satisfaction (mean)	GDP growth (5-year)	334	0.02	0.16	0.34	0.02	0.52
Happiness (mean)	Freedom of choice	333	0.39	0.02	0.22	0.27	0.51
Happiness (mean)	Importance of God	330	0.01	0.05	0.31	0.03	0.39
Happiness (mean)	Tolerance (homosexuality)	325	0.09	0.02	0.31	0.07	0.40
Happiness (mean)	Importance of democracy	216	0.03	0.02	0.28	−0.03	0.27
Happiness (mean)	GDP growth (5-year)	334	0.01	0.03	0.31	0.00	0.34

Note: Country-year averages of WVS variables: perceived freedom of choice and control (1–10), importance of God in one’s life (1–10), justifiability of homosexuality (1–10), importance of living in a democracy (1–10, waves 5–7), and average annual growth of real GDP per capita over the previous five years. Shapley decomposition of the R^2 of the regression on log GDP per capita, region dummies and the other variable.

Table A5: Within-country relation between national well-being and $\log_{10}$ GDP per capita (country fixed effects, WVS)

Indicator	Slope	s.e.	p -value	Within R^2	Obs.
Happy	0.286	0.043	0.000	0.281	312
Very Happy	0.207	0.048	0.000	0.146	312
Very Unhappy	−0.039	0.010	0.000	0.044	312
V. Happy – V. Unhappy	0.246	0.054	0.000	0.160	312
Happiness (mean)	1.063	0.180	0.000	0.245	312
Satisfaction (mean)	2.313	0.420	0.000	0.344	312
Satisfied	0.440	0.076	0.000	0.347	312
Happy + Satisfied	0.366	0.058	0.000	0.389	310

Note: Countries surveyed at least twice. Standard errors clustered by country.

Table A6: Countries most often ranked happiest (WVS, 8 indicators $\times$ 8 samples: each wave and all waves combined)

Country	Occurrences as happiest	Region
Switzerland	10	Western
Mexico	9	Latin America
Kyrgyzstan	6	Asia
Australia	5	Western
Netherlands	4	Western
Nigeria	4	Africa
Puerto Rico	4	Latin America
Colombia	3	Latin America
Finland	3	Western
Vietnam	3	Asia

Figure A1: Share of *Very happy* people vs. GDP per capita (PPP), WVS/EVS 1981–2023

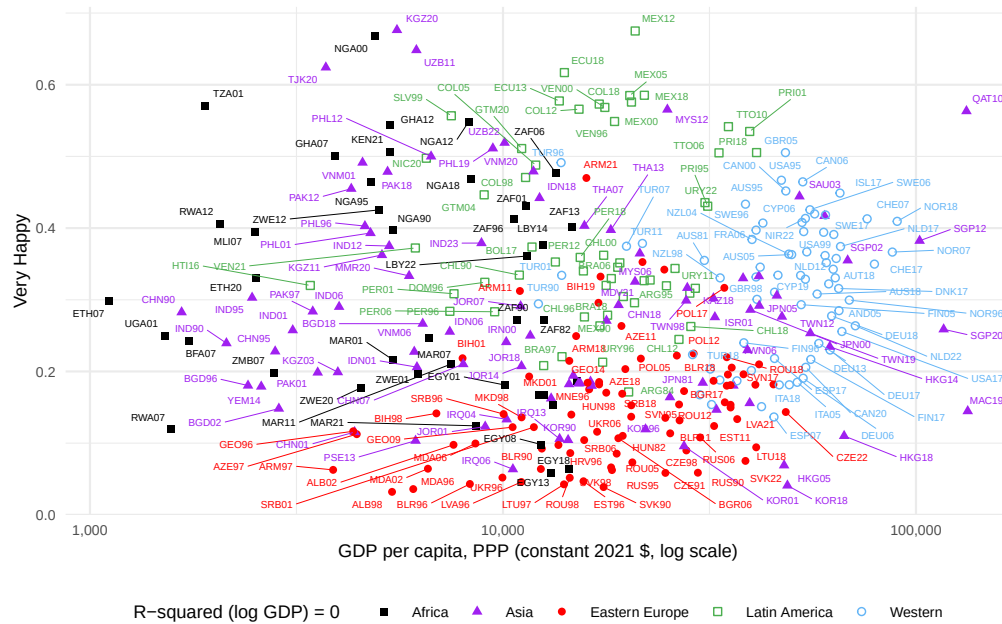


Figure A2: Mean ladder vs. GDP per capita (PPP), Gallup, last observation per country

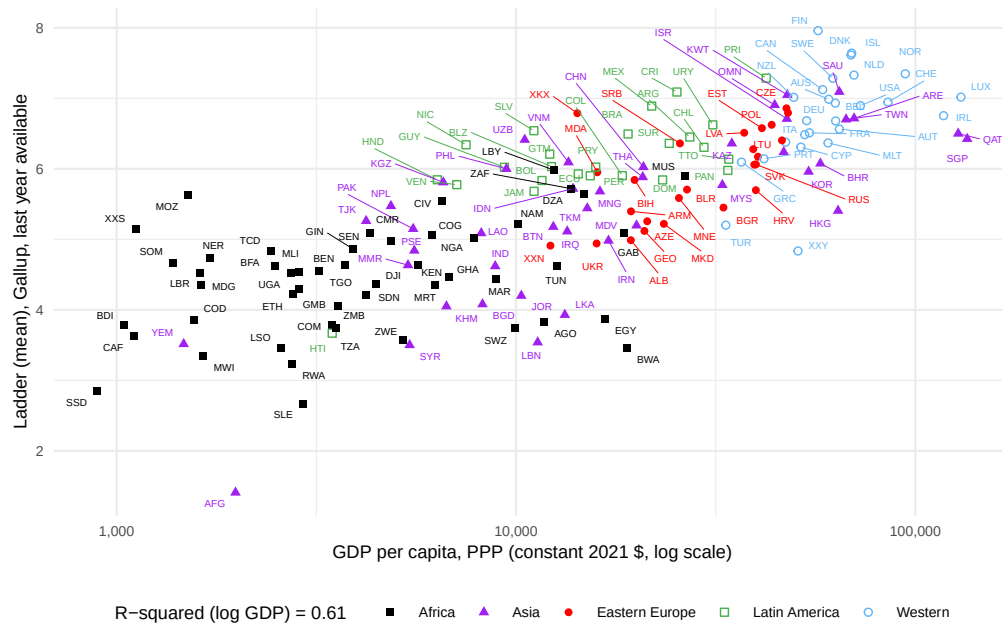


Figure A3: *Happy* vs. GDP per capita (PPP), WVS/EVS 1981–2023

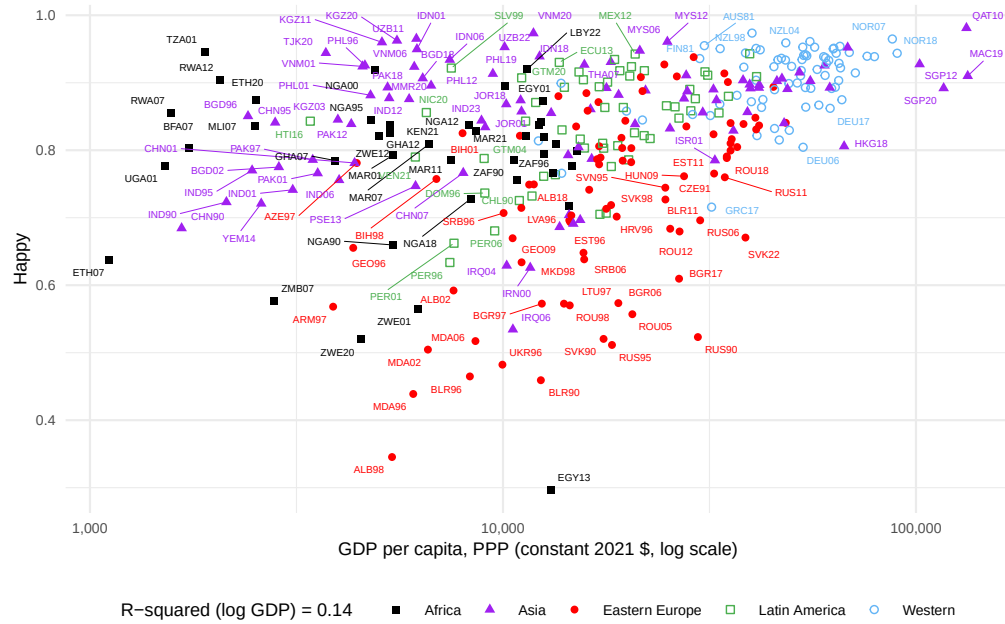


Figure A4: *Very Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023

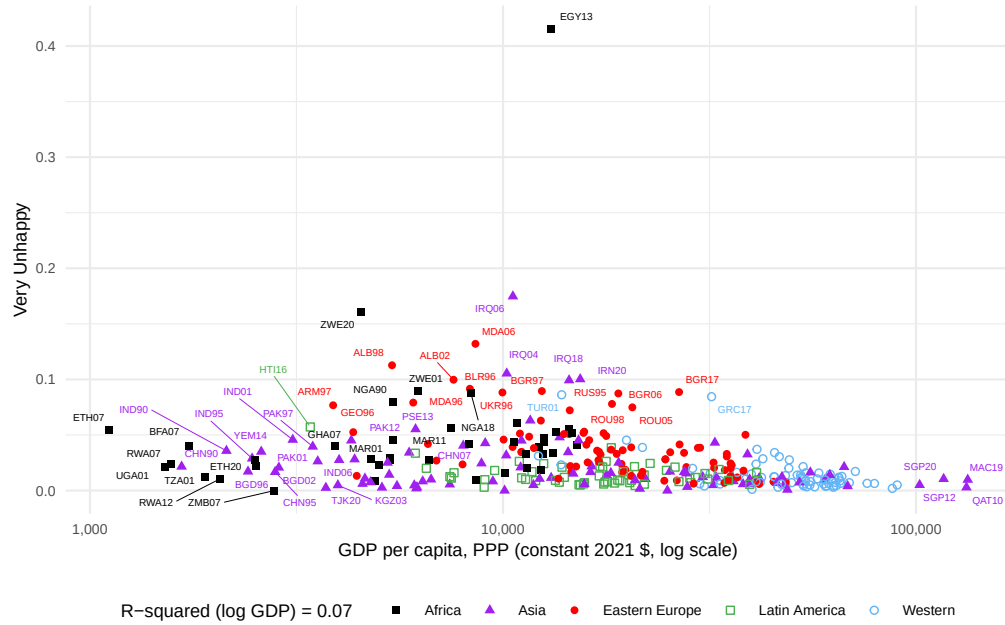


Figure A5: *V. Happy* – *V. Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023

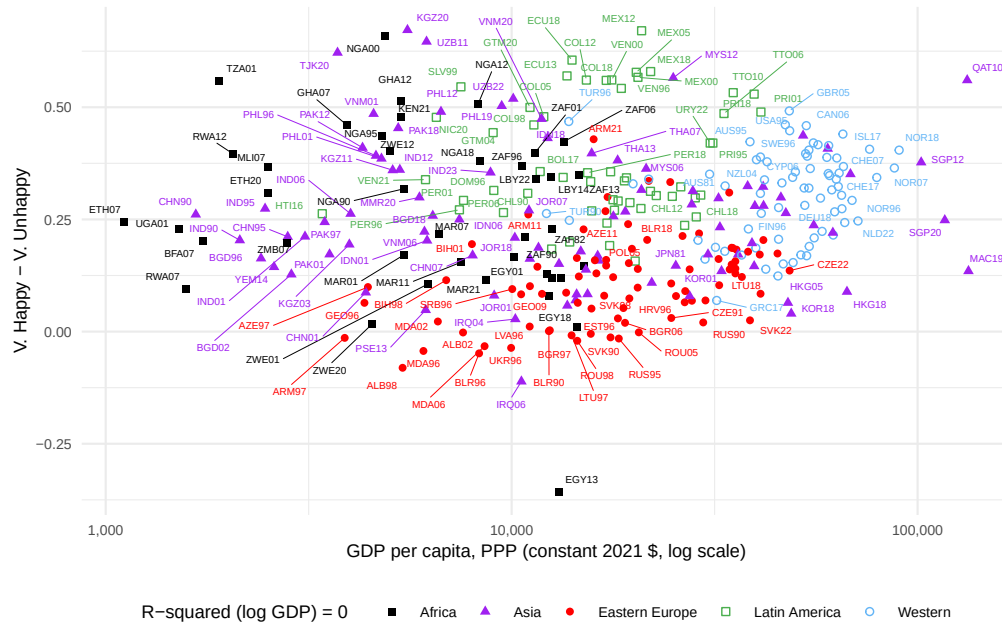
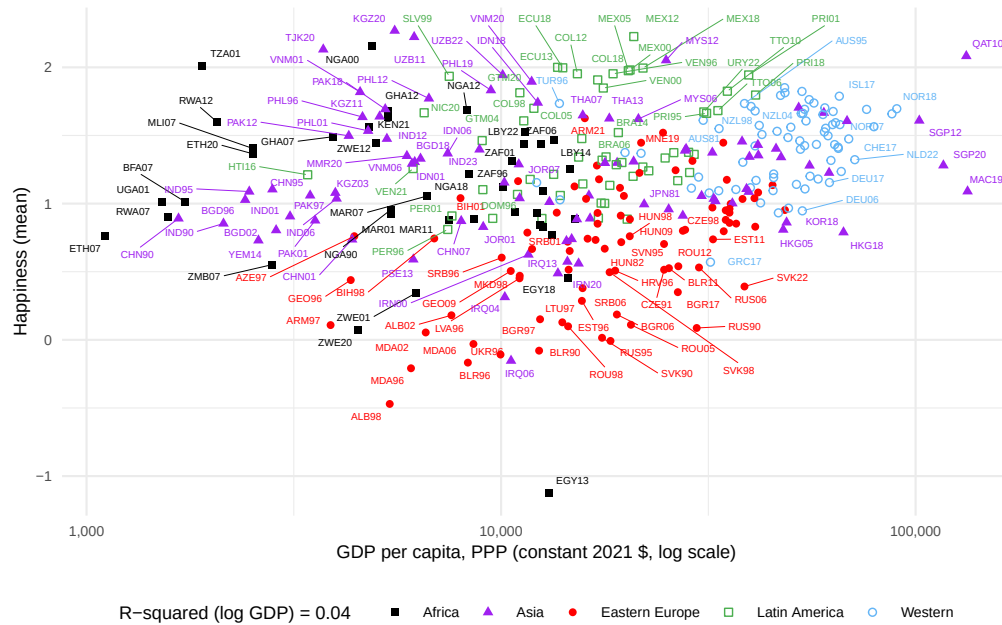


Figure A6: *Happiness (mean)* vs. GDP per capita (PPP), WVS/EVS 1981–2023

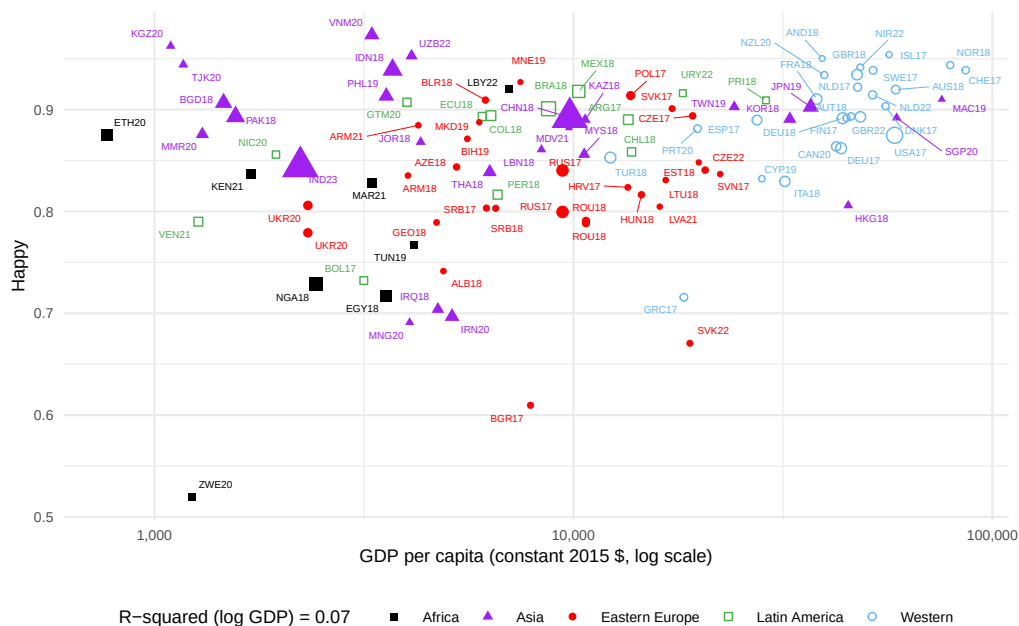


R-squared (log GDP) = 0.32

Legend:

- Africa (Black square)
- Asia (Purple triangle)
- Eastern Europe (Red circle)
- Latin America (Green square)
- Western (Blue circle)

Figure A9: *Happy* vs. GDP per capita (nominal), WVS/EVS wave 7 (2017–2023), weighted by population



Note: Point size and regression weights proportional to population.